

Since the beginning of the year, Intel and AMD have been releasing speedier iterations of their processors with Intel's fastest Pentium 4 CPU now running at a whopping 2.4 GHz and AMD claiming its 1.73 GHz baby is faster than Intel's 2 GHz+ processors.

The fanatics from both camps have screamed their lungs out at online forums, group discussions, and press presentations

and the flame wars have pursued. Some say that AMD's PR (Performance Rating) scheme is no good, while others argue that clock frequency makes no difference. The marketing teams of both these chip giants have pushed and shoved every known jargon at the end user, from NetBurst to QuantiSpeed architectures and SSE2 to 3DNow! 'Professional'—that's enough to confuse even the technogeeks.

But the time has come to wipe the slate clean, to take a fresh look at what each company has to offer and verify their outlandish claims with a slew of benchmarks—both real world and next generation—that are relevant to all computing usage models. Ultimately, it's pure performance that counts. May the best processor win the fight!

Professional Applications

Finally, the Athlon XP has something to boast about. It beats the Pentium 4 hollow in these tests. High-end imaging and 3D authoring applications are FPU dependent and Athlon XP's FPU strength is unparalleled, which allows it to beat the Pentium 4 by hefty margins. Since the introduction of the original Athlon (K7), AMD processors have had the upper hand when it came to raw FPU muscle. The Athlon XP builds upon the same core and hurts the Pentium 4, which can do little without proper SSE2 support.

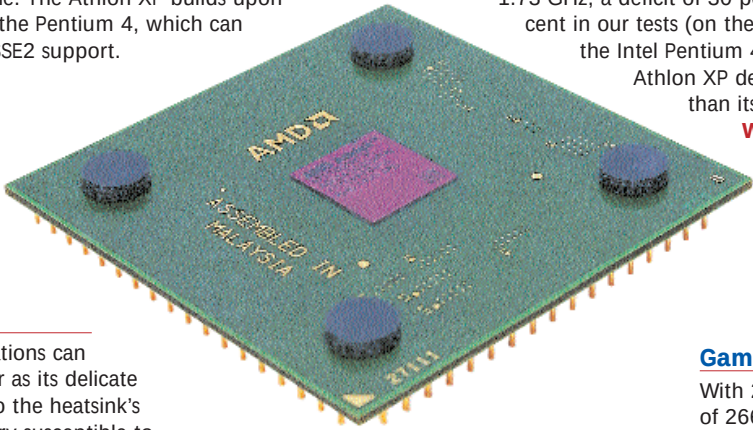
Winner: Athlon XP

Business Performance and Office Productivity

In the past, Athlon XP has managed to prove its mettle in these tests even when pitted against the fastest Intel processors. Despite its higher IPC (instructions per clock or work done per cycle), the Athlon XP 2100+ is slower by a whopping 667 MHz when compared to the mighty Pentium 4 2.4 GHz and simply cannot overcome a difference of this magnitude. Nevertheless, its scores are quite impressive when you consider that it runs at 1.73 GHz, a deficit of 30 per cent, and only loses out by 10 per cent in our tests (on the other hand, it does manage to beat the Intel Pentium 4 2.2 GHz!). This proves that the

Athlon XP deserves a faster performance rating than its pure frequency speed.

Winner: Pentium 4



Durability

Improper heatsink installations can smash the AMD processor as its delicate core is directly exposed to the heatsink's pressure. This makes it very susceptible to cracking or chipping if the heatsink is incorrectly mounted. And even though Athlon XP now comes with some form of CPU overheat protection, it's nowhere in comparison to what Intel offers.

Winner: Pentium 4

Price

Who can argue with a price difference of Rs 15,000? There's just no contest here. AMD has always positioned its processors as true value for money propositions. Pricing has always been AMD's trump card and they've been winning this battle hands down. The street price of an Athlon XP 2100+ is Rs 20,000 which is way cheaper than its Intel counterpart. The Athlon XP is your true price to performance champion as it offers top-of-the-line performance without burning a large hole in your pocket.

Winner: Athlon XP

Gaming and Multimedia

With 256 KB of L2 cache and an FSB speed of 266 MHz, the Athlon XP can do little to overtake the Pentium 4 in this department. Quake III Arena has never been kind to the AMD processor and if you're interested in purchasing the fastest possible CPU to go with your spanking new GeForce4, then don't even bother with the Athlon XP, which lags behind its Intel counterpart by a huge 23 per cent! But the Athlon makes up for its lackadaisical performance in Direct3D games, losing by a mere 3 per cent in 3DMark2001 SE. Sadly, it blows a lot of hot steam, recording temperatures in excess of 50 degree Celsius while running our tests.

Winner: Pentium 4

Test	
Pentium 4 2.4 GHz	Athlon XP 2100+
71.5	63.1
90.3	82.5
7720	7495
375.8	304.1
9483	9376
11511	10980
97	95
58.13	59.03
555	376

Make a Choice

Clock to clock, the Athlon XP is a faster processor. If there was an Athlon XP 2400+, we're quite certain that it would have won hands down. But the reality is that Intel has been steadily releasing faster processors and the Pentium 4 2.4 GHz is the current performance king. The Athlon

XP 2100+ performs admirably but just cannot make up for the raw speed deficit of 667 MHz. Without a ramp up to 0.13 micron technology, it seems highly unlikely that AMD will be able to cover this speed gap anytime soon.

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